



02.07.2026
Tag 9

Azure for Students

Learning Path
Networks

→ ^{meine} Subscription 100\$ 1 Jahr

Von KI erstellt, von Menschen kontrolliert.

Part 1 ✓

Part 2 ✓

Part 3 ✓

move

Mein Tenant

Der Dritte Mann

GitHub Repo <https://github.com/www42/a5>

Part 4

AZ-104

Learning Paths:

- Manage identities and governance in Azure
- Prerequisites for Azure administrators – CloudShell and templates
- Configure and manage virtual networks for Azure administrators
- Implement and manage storage in Azure
- Deploy and manage Azure compute resources
- Monitor and back up Azure resources

What happened yesterday



Azure Copilot (kostenlos!)

Gemini
Gemini

GPT 5.5
Fables

What are agents?

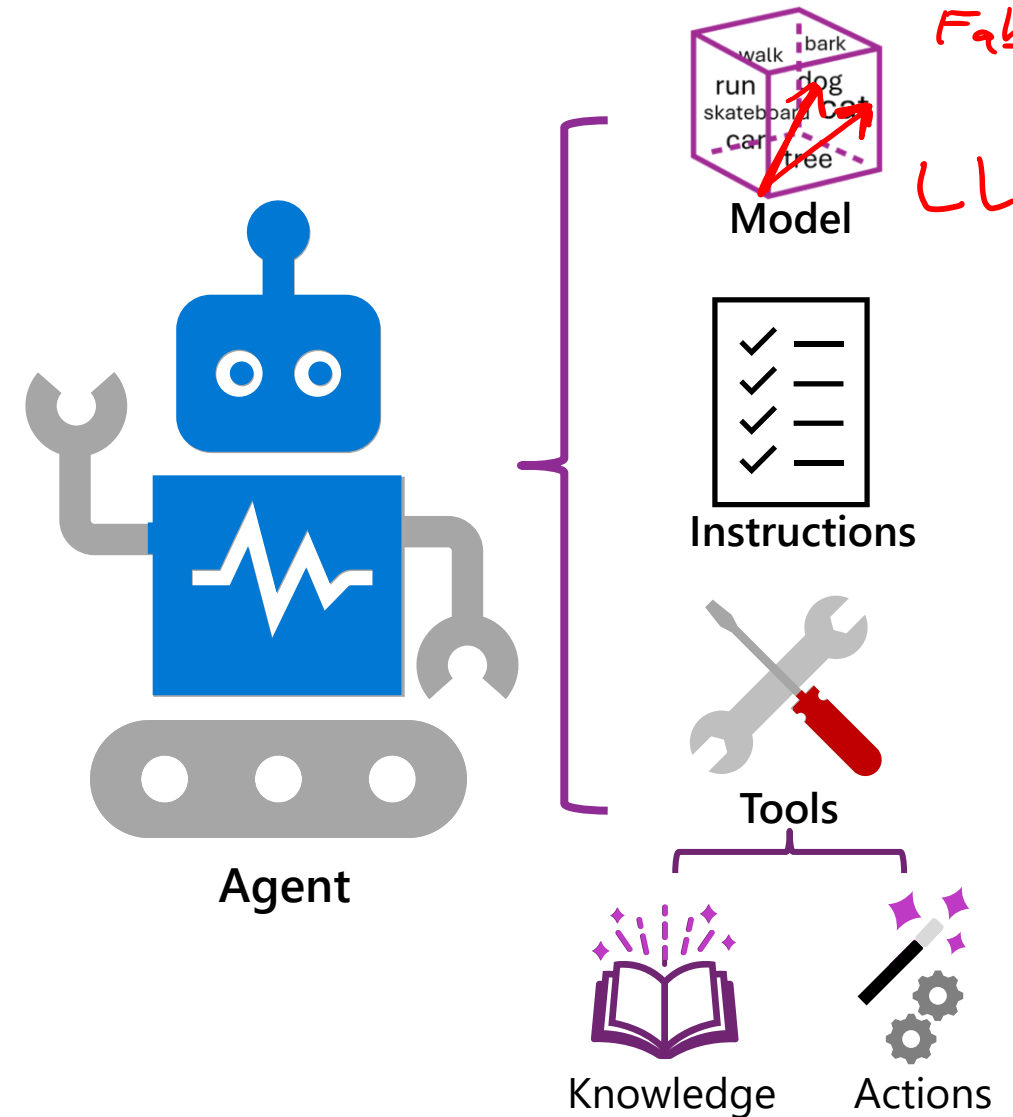
Autonomous, action-capable AI assistants

Agents use:

- A **model** for language understanding and reasoning
- **Instructions** to define behavior and scope
- **Tools** they can use to:
 - Find **knowledge** to contextualize their work
 - Perform **actions** to automate tasks

Agents collaborate with:

- Human users, to assist with tasks and processes
- Other agents, to create workflows of specialized agents, each with their own responsibilities



Part 1

GfN Tenant



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GfN GmbH (Edu)



Subscription 100\$

Part 2

Mein Tenant



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No Subscription

move

Azure for Students

Schicksal göttinnen

Atropos

Lachesis

Klotho

schneidet ab

Länge

spinnt

AZ-104

 Configure and manage virtual networks for Azure administrators

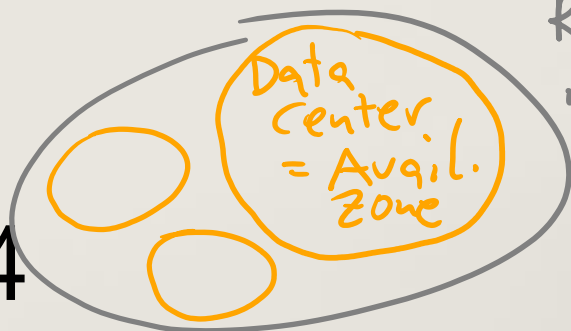




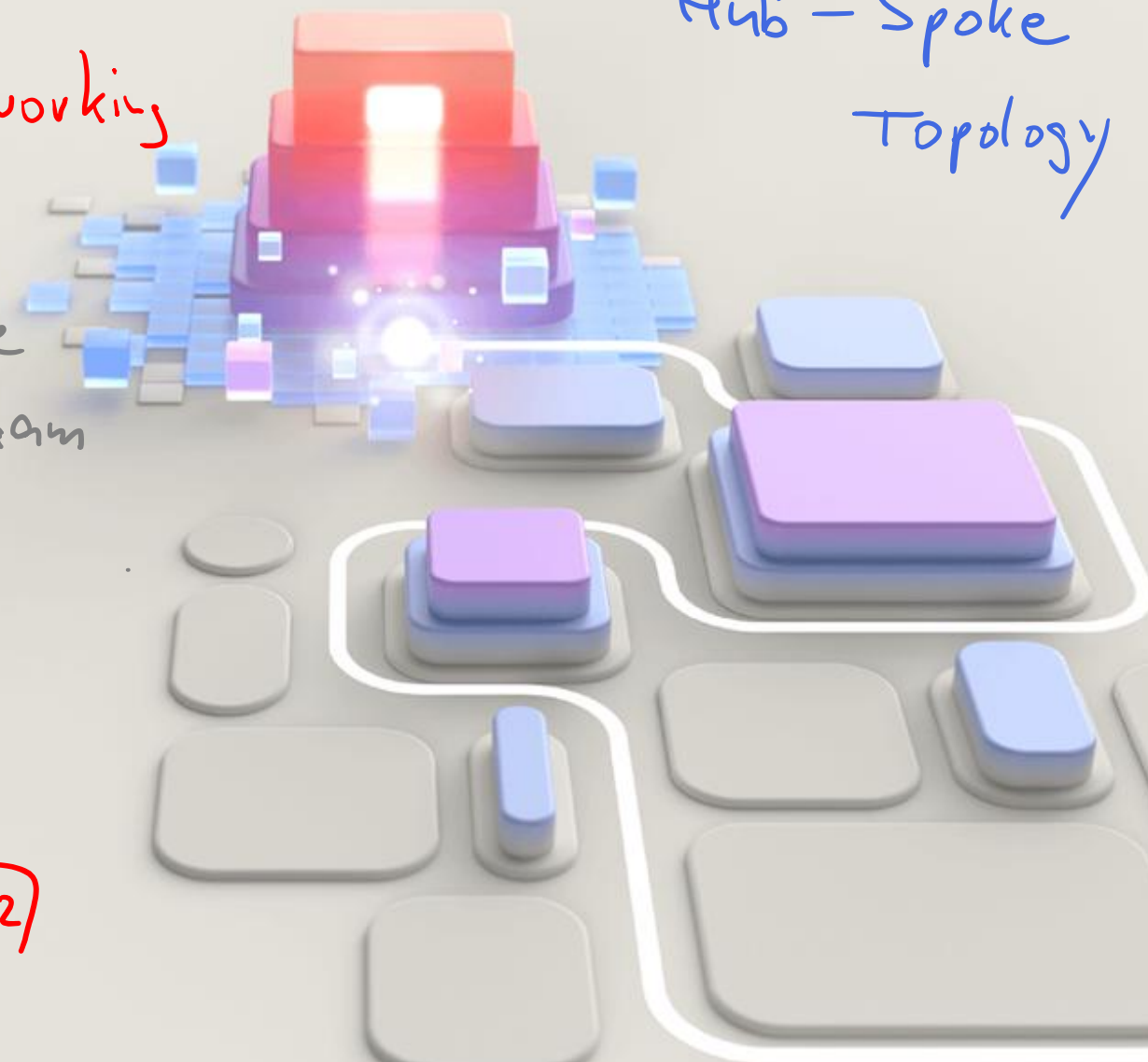
SDN Software Defined Networking

Hub-Spoke
Topology

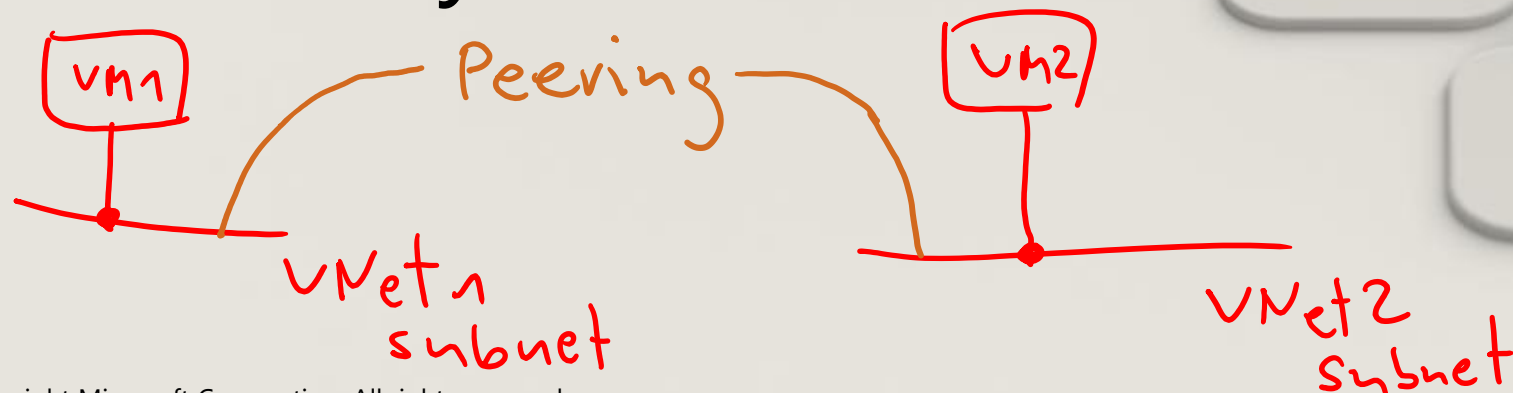
AZ-104



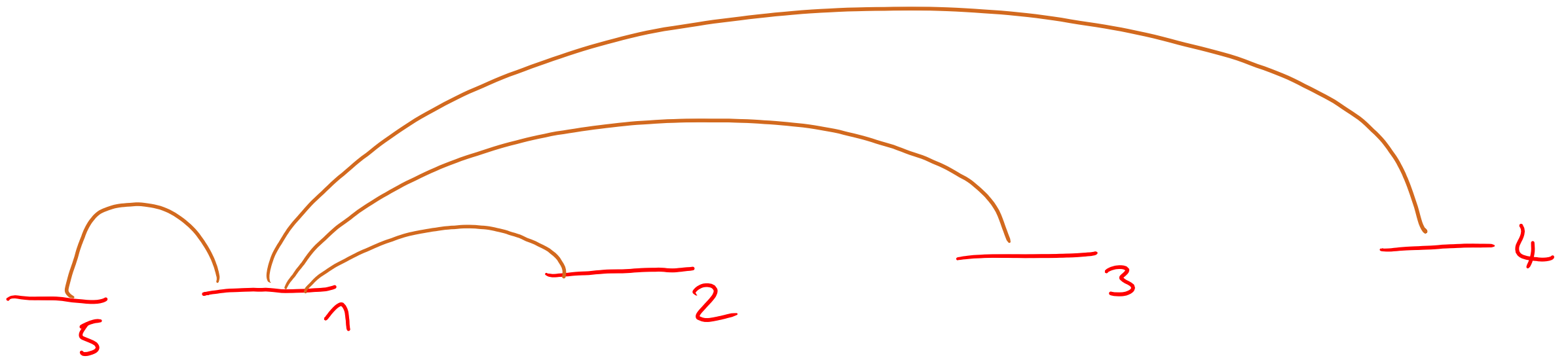
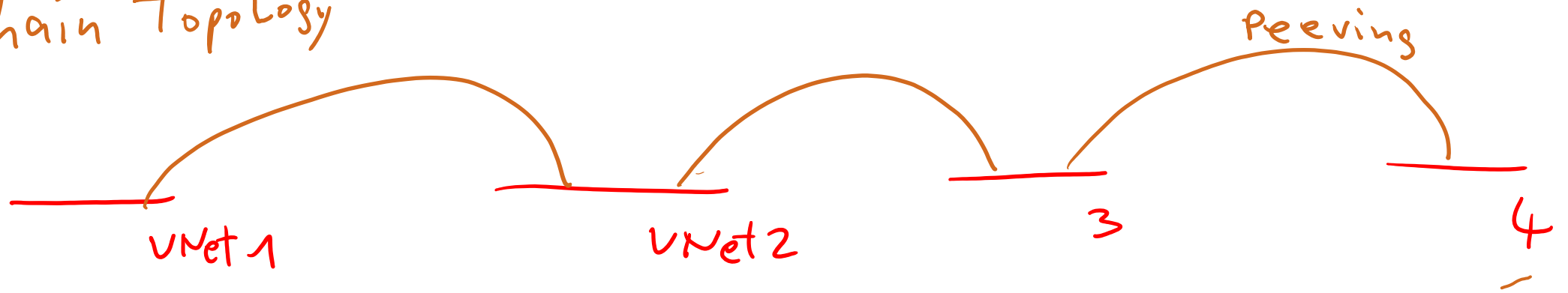
Region
westeurope
Amsterdam



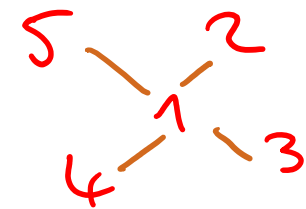
Administer Intersite Connectivity



Chain Topology



Hub-Spoke Topology

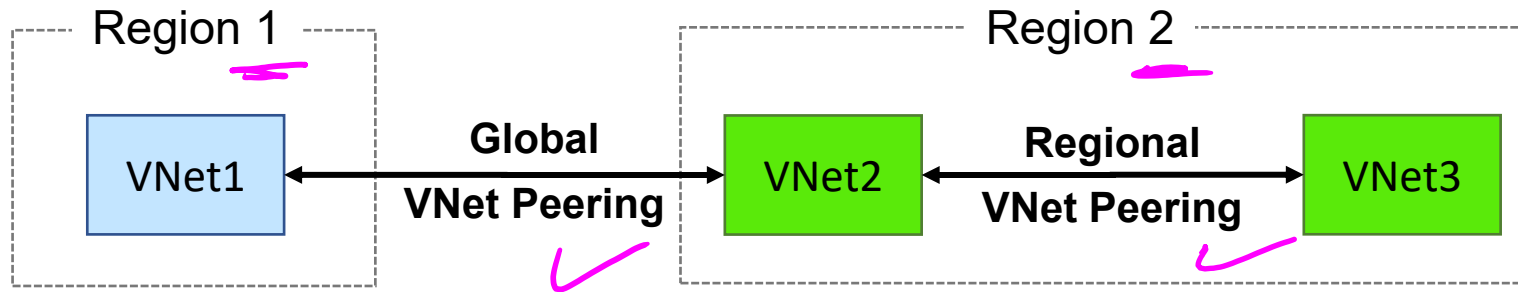


Configure VNet Peering



Determine VNet Peering Uses

Peering is not encrypted.
Encryption? → VPN
(Virtual Network Gateway)



- Two types of peering: Global and Regional
- Connects two Azure virtual networks – you can peer across subscriptions and tenants
- Peered networks use the Azure backbone for privacy and isolation
- Easy to setup, seamless data transfer, and great performance

GW = Default Gateway

BGP dynamic Routing

Determine Gateway Transit and Connectivity Needs

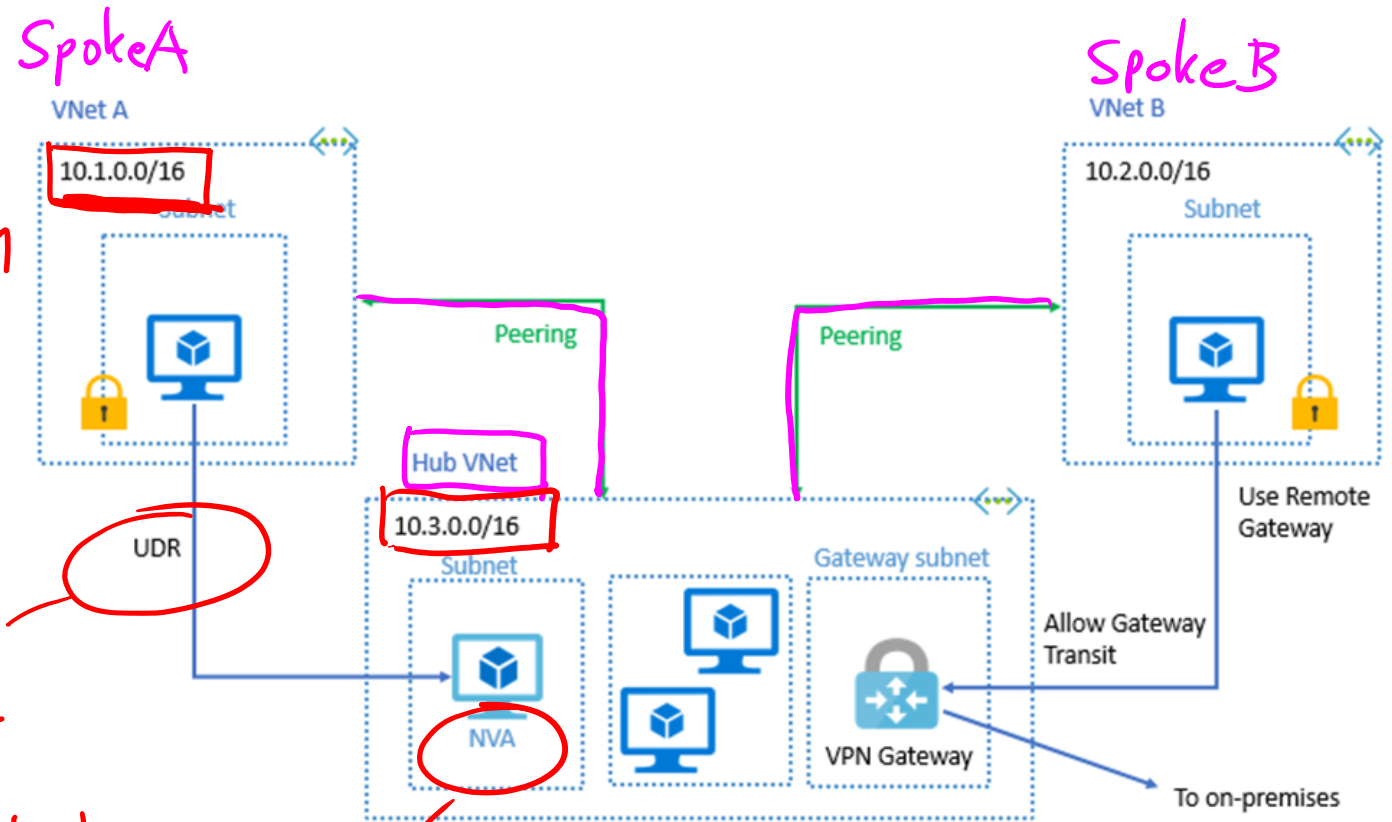
- Gateway transit allows peered virtual networks to share the gateway and get access to resources
- No VPN gateway is required in the peered spoke virtual network
- Default VNet peering provides full connectivity

10.1.0.1
Default GW

Static Routing

User Defined Route

Network Virtual Appliance



IP address spaces of connected networks can't overlap

Create VNet Peering

- Allow virtual network traffic
- Allow forwarded traffic
- Allow gateway or route server traffic
- Status should show "connected"

Remote virtual network summary

Peering link name *

Remote virtual network peering settings

Allow 'vnet-2' to access 'vnet-1' ⓘ ☒

Allow 'vnet-2' to receive forwarded traffic from 'vnet-1' ⓘ ☒

Allow gateway or route server in 'vnet-2' to forward traffic to 'vnet-1' ⓘ ☐

Enable 'vnet-2' to use 'vnet-1's' remote gateway or route server ⓘ ☐

Local virtual network summary

Peering link name *

Local virtual network peering settings

Allow 'vnet-1' to access 'vnet-2' ⓘ ☒

Allow 'vnet-1' to receive forwarded traffic from 'vnet-2' ⓘ ☒

Allow gateway or route server in 'vnet-1' to forward traffic to 'vnet-2' ⓘ ☐

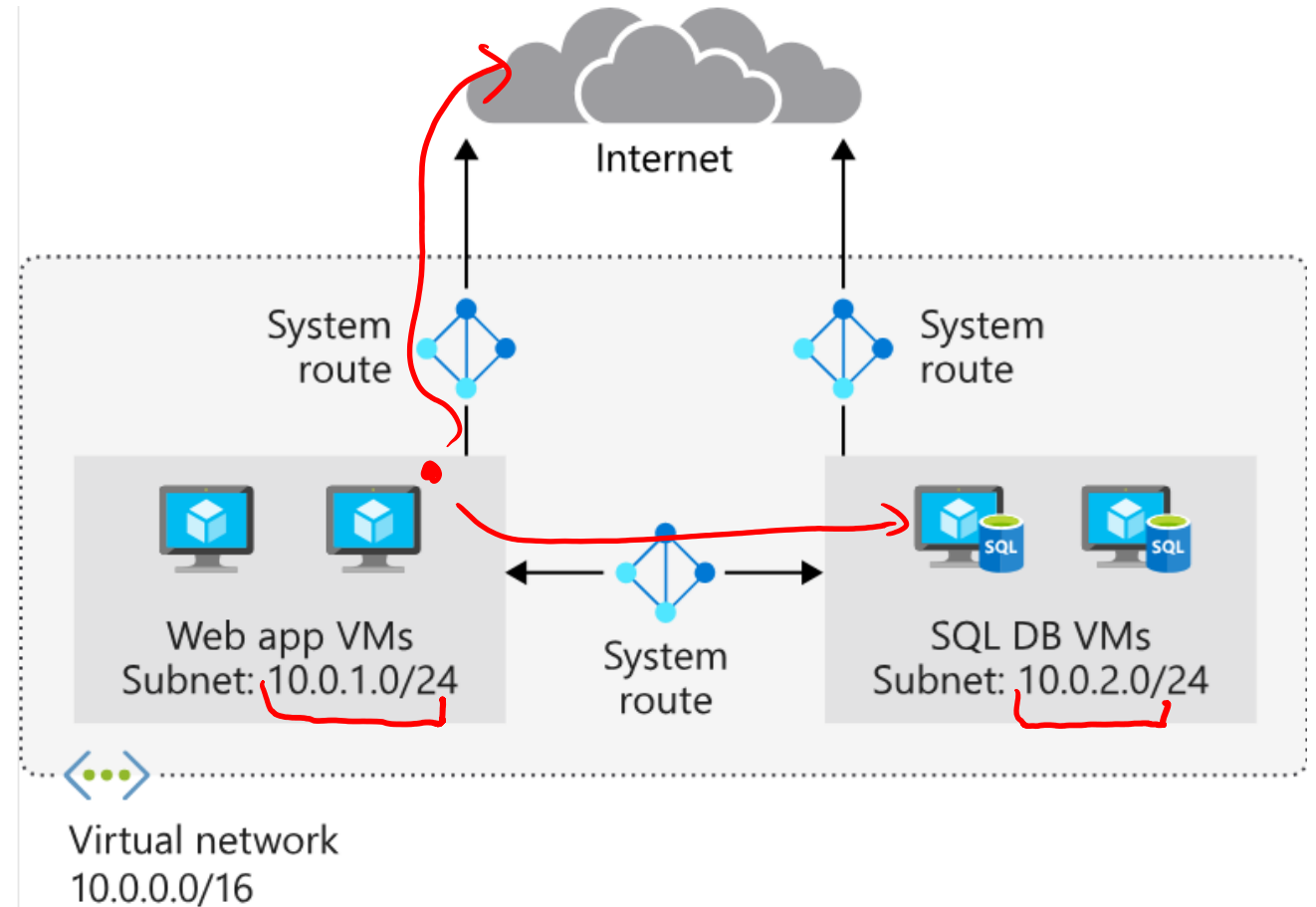
Enable 'vnet-1' to use 'vnet-2's' remote gateway or route server ⓘ ☐

Manage and control traffic flow with routes



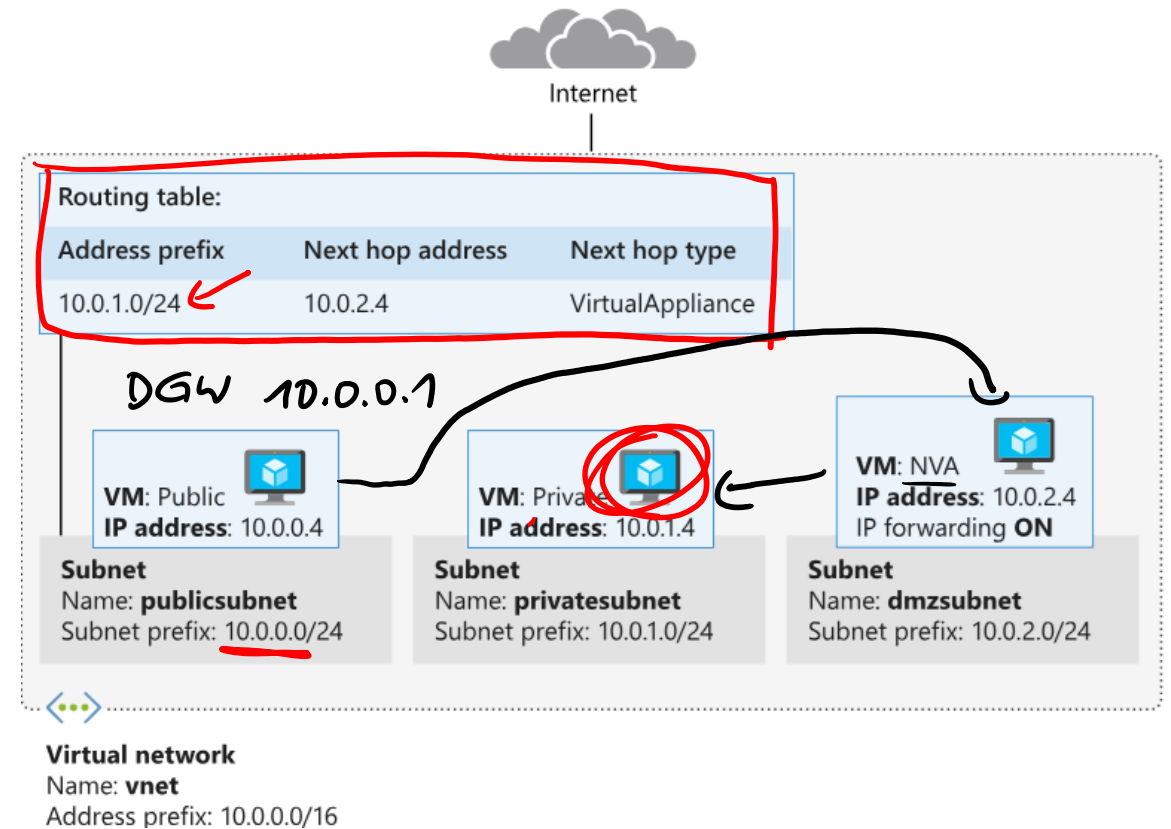
What are System Routes?

- Pre-defined routes automatically added to every subnet's route table by Azure
- Ensures fundamental connectivity (e.g., VNet-to-VNet, VNet-to-Internet, VNet-to-Azure services)
- **Examples:** Routes for VNet peering, local traffic, and the 0.0.0.0/0 internet route
- You can't delete system routes, but you *can* override with custom routes (UDRs)



What are User-Defined Routes (UDRs)?

- UDRs override the default system routes
- Traffic can be directed to virtual appliance, Virtual Network Gateway, Virtual Network, Internet, or None (drop it)
- Border Gateway Protocol (BGP) exchanges routes between Azure and on-premises networks
- You can specify a service tag as the address prefix for a user-defined route instead of an explicit IP range



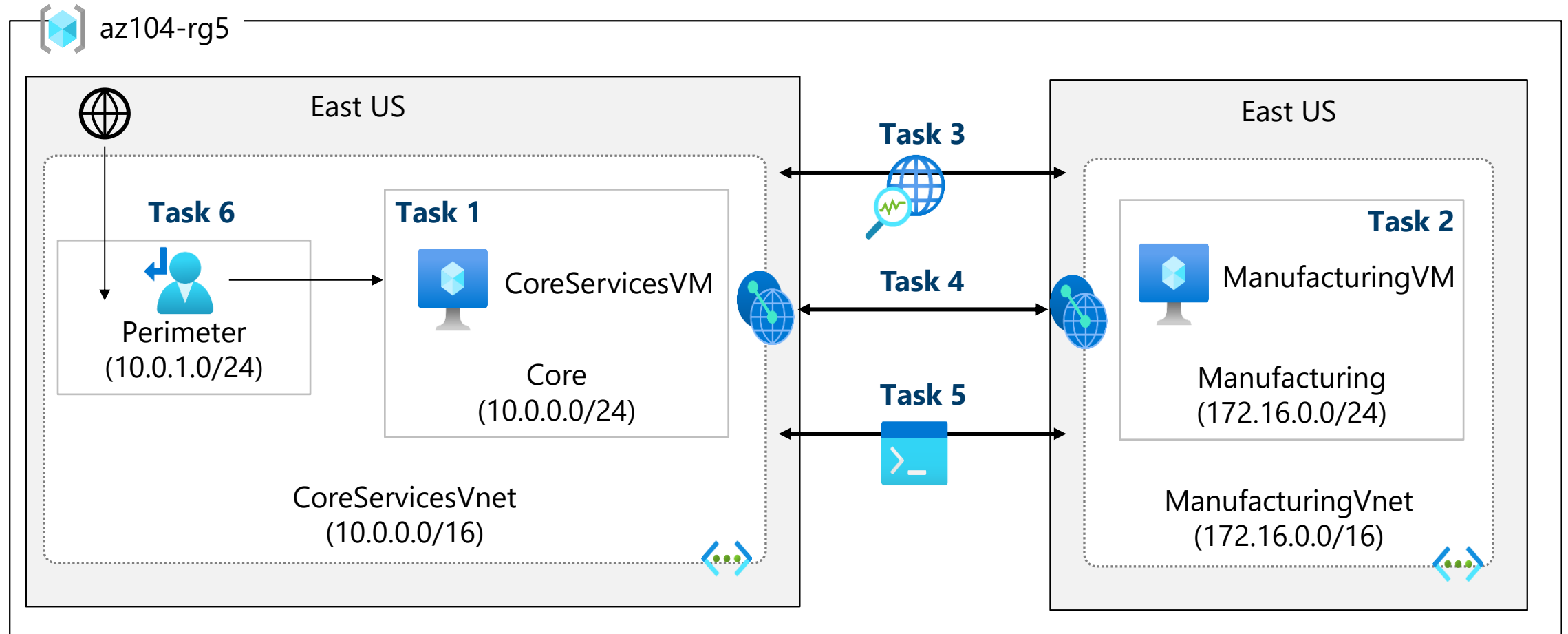
Labs



Lab 05 - Implement Intersite Connectivity



Lab 05 – Architecture diagram



End of presentation



